

Prof. Gayathri Bharathan

ASSISTANT PROFESSOR

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PROFESSIONAL SUMMARY

Dynamic academic leader with more than seven years of experience in premier institutions, including as Assistant Professor at IIT Delhi. Proven track record in driving institutional strategy, research development, academic quality enhancement, and student engagement. With proven experience in establishing and managing advanced research facilities, securing funding, and building global academic partnerships, I aim to bring innovation, quality education, and impactful scholarship to support the educational institute's vision and growth.

EDUCATION

- Ph.D. in Physics (Photonics) – Macquarie University, Australia (2018–2021)
- Master of Research (M.Res.) – Macquarie University, Australia (2017–2018)

WORK EXPERIENCE

- Assistant Professor, Indian Institute of Technology (IIT) Delhi, India (April 2023 – Till now)
- Guest Researcher, University of Tokyo (Research collaboration visit from IIT Delhi), Tokyo, Japan (Feb 2025 – April 2025)
- Research Scientist (Research collaboration visit from IIT Delhi), Macquarie University, Sydney, Australia (Aug 2024 – Till now)
- Research Scientist (Post-doctoral Fellow), Photonic Systems Laboratory, Department of Electrical Engineering, EPFL, Switzerland (April 2021 – Feb 2023)
- Teaching Assistant, Faculty of Science and Engineering, Macquarie University, Australia (2018–2020)
- Teaching Assistant, Advanced Photonics Laboratory, Dept. of Electrical Engineering, Indian Institute of Technology (IIT) Madras, India (2014–2016)

PROFESSIONAL EXPERIENCE GAINED

- Conducted independent research projects in the field of laser science, published in high-impact journals, and presented the research work at various conferences.
- Collaborated with interdisciplinary teams to identify and write grant applications and secure funding for research projects involving laser technology.
- Managed and negotiated research contracts with various government and private sector entities.
- Provided research support services to academic colleagues, including literature reviews, data analysis, and manuscript preparation.
- Managed and tracked multiple research projects concurrently, ensuring adherence to timelines, deliverables, and budgets.
- Facilitated communication and collaboration between faculty members and external partners to foster research innovation
- Assisted in organizing research seminars, workshops, and conferences to promote knowledge sharing within the academic community.

SKILLS

- **Teaching:** Ability to explain complex concepts clearly and make learning engaging for students.
- **Student Mentorship:** Experience in mentoring and guiding students to achieve academic and career success.
- **Research Income Identification:** Expertise in identifying and securing funding opportunities across diverse fields, including science and engineering, with a focus on emerging technologies.
- **Research Strategy Development:** Strong background in assisting with the development and implementation of research strategies aligned with faculty priorities.
- **Stakeholder Engagement:** Experienced in liaising with academic staff, external funding bodies, and industry partners to promote collaborative research projects.
- **Collaboration:** Proven ability to engage with internal and external stakeholders to foster collaborative and multidisciplinary research.
- **Data Management & Communication:** Skilled in collecting, analyzing, and communicating research data to inform decision making and strategy adjustments.
- **Project Management:** Effective in managing multiple research projects, ensuring timely delivery and successful execution of strategic
- **Administrative Operations:** Expertise in managing research administration, including internal funding schemes, milestone tracking, and compliance reporting.

PUBLICATION LIST

Book Chapter

- Fuerbach, G. Bharathan, M. Ams: "Direct-written gratings in soft glass fibres", invited chapter in Mid-Infrared Fibre Photonics, (R. Vallee, M. Bernier, S. Jackson, eds.), Elsevier, Netherlands (2021).

Journal Articles

- T. T. Ha, G. Bharathan, and A. Fuerbach. Chalcogenide-glass Waveguide Arrays for Mid-Infrared Mode-Locked Fiber Lasers. *Journal of Lightwave Technology* (2025).
- Trong Thuy Ha, Gayathri Bharathan, and Alex Fuerbach, "Linear and nonlinear optical properties of femtosecond laser inscribed waveguides into GLS glass," *Opt. Express* 32, 42722-42731 (2024).
- T Toney Fernandez, Luyi Xu, Simon Gross, Gayathri Bharathan, Martin Ams, Karen Privat, Michael J Withford, Alex Fuerbach: "Physical mechanisms of femtosecond laser induced refractive index change in direct-written mid-infrared fiber Bragg gratings," *APL Photonics*, 9, 110802 (2024).
- Moritz Bartnick, Gayathri Bharathan, Thorsten Albert Goebel, Ria G. Kramer, Stefan Nolte, and Camille-Sophie Bres: "Wavelength-stabilized figure-of-9 thulium-doped all-fiber laser emitting 560 fs pulses," *Laser Phys*, 34 025101 (2023).
- Moritz Bartnick, Gayathri Bharathan, Thorsten Albert Goebel, Ria G. Kramer, Stefan Nolte, and Camille-Sophie Bres: "Wavelength-stabilized tunable mode-locked thulium-doped fiber laser beyond 2 μm ", *Opt. Lett.*, 47, 2085-2088 (2022).
- Gayathri Bharathan, Luyi Xu, Xiantao Jiang, Han Zhang, Feng Chen, Ziqi Li, and Alex Fuerbach, "MXene and PtSe₂ saturable absorbers for all-fibre ultrafast mid-infrared lasers," *Opt. Mater. Express* 11, 1898-1906 (2021).
- Matthew Majewski, Gayathri Bharathan, Alex Fuerbach, and Stuart D. Jackson, "Long wavelength operation of a dysprosium fiber laser for polymer processing", *Opt. Lett.* 46, 600-603 (2021).
- Gayathri Bharathan, Toney Teddy Fernandez, Martin Ams, Jean-Yves Carrée, Samuel Poulain, Marcel Poulain and Alex Fuerbach, "Femtosecond laser direct-written fiber Bragg gratings with high reflectivity and low loss at wavelengths beyond 4 μm ", Accepted in *Opt. Lett.* (July 2020).
- Alex Fuerbach, Gayathri Bharathan and Martin Ams, "Grating inscription into fluoride fibers: A review", *IEEE Photonics Journal*. 11 (2019).
- Gayathri Bharathan, Toney Teddy Fernandez, Martin Ams, Robert I. Woodward, Darren D. Hudson and Alex Fuerbach, "Optimized laser written ZBLAN fiber Bragg gratings with high reflectivity and low loss", *Opt. Lett.* 44, 423-426 (2019).
- Gayathri Bharathan, Darren D. Hudson, Robert I. Woodward, Stuart D. Jackson, and Alex Fuerbach, "In-fiber polarizer based on a 45° tilted fluoride fiber Bragg grating for mid-infrared fiber laser technology", *Optics Continuum* 1, 56-63 (2018).

- Thorsten A. Goebel, Gayathri Bharathan, Martin Ams, Maximilian Heck, Ria G. Kramer, Christian Matzdorf, Daniel Richter, Malte P. Siemsi, Alex Fuerbach and Stefan Nolte "Realization of aperiodic fiber Bragg gratings with ultrashort laser pulses and the line-by-line technique", *Opt. Lett.* 43, 3794-3797 (2018).
- R. I. Woodward, M. R. Majewski, G. Bharathan, D. D. Hudson, A. Fuerbach, and S. D. Jackson, "Watt-level dysprosium fiber laser at 3.15 μm with 73% slope efficiency," *Opt. Lett.* 43, 1471-1474 (2018).
- Gayathri Bharathan, Robert I. Woodward, Martin Ams, Darren D. Hudson, Stuart D. Jackson, and Alex Fuerbach, "Direct inscription of Bragg gratings into coated fluoride fibers for widely tunable and robust mid-infrared lasers", *Opt. Express* 25, 30013-30019 (2017).

Conference Articles

- A. Fuerbach, G. Bharathan, T. Fernandez. "Femtosecond laser induced refractive index changes in fluoride glasses", invited talk, 10th Conference on Applications of Femtosecond Lasers in Materials Science (FemtoMat 2025), Austria, 2025.
- A. Fuerbach, G. Bharathan, T. T. Ha, Luyi Xu, T. Fernandez. "Femtosecond Laser Written Components for Mid-Infrared Photonics", invited talk, CLEO US, 2025.
- W. Hussain, T. T. Ha, G. Bharathan, L. Xu and A. Fuerbach. Fabrication and characterization of high-index contrast waveguides in Suprasil 3001 for 3.2 μm wave-length applications. Accepted as an oral presentation at the European Conference on Lasers and Electro-Optics (CLEO), Munich, Germany (2025).
- T. T. Ha, G. Bharathan, L. Xu and A. Fuerbach. GLS Waveguide Array: A Novel Saturable Absorber for Mid-Infrared Mode-Locked Fiber Lasers. Accepted as an oral presentation at the European Conference on Lasers and Electro-Optics (CLEO), Munich, Germany (2025).
- T. T. Ha, G. Bharathan, and A. Fuerbach. GLS waveguide array: A novel saturable absorber for mid-infrared pulsed fibre lasers. The 2024 Australia and New Zealand Conference on Optics and Photonics (ANZCOP) and Australian Institute of Physics (AIP) Summer Meeting, Melbourne Convention Exhibition Centre, Melbourne (2024).
- Moritz Bartnick, Gayathri Bharathan, and Camille-Sophie Bres: Polarization-multiplexed thulium-doped fiber laser for free-running dual-comb generation, SPIE LASE, 2023, San Francisco, California, USA.
- Moritz Bartnick, Gayathri Bharathan, Thorsten Albert Goebel, Ria G. Kramer, Stefan Nolte, and Camille-Sophie Bres: Tunable wavelength-stabilized mode-locked thulium-doped fiber laser beyond 2000 nm, SPIE Photonics Europe, 2022, Strasbourg, France.
- Moritz Bartnick, Gayathri Bharathan, Laure Lago-Rached, Arnaud Mussot, and Camille-Sophie Bres: "Wavelength-stabilized mode-locked thulium-doped fiber laser operating between 1958 and 2008 nm", CLEO US 2022, California, USA.
- Moritz Bartnick, Gayathri Bharathan, Thorsten Albert Goebel, Ria G. Kramer, Stefan Nolte, and Camille-Sophie Bres: "Tunable wavelength-stabilized mode-locked thulium-doped fiber laser beyond 2000 nm," *Proc. SPIE* 12142, Fiber Lasers and Glass Photonics: Materials through Applications III, 121420N (2022).
- Moritz Bartnick, Gayathri Bharathan, Thorsten Albert Goebel, Ria G. Kramer, Stefan Nolte, and Camille-Sophie Bres: "Wavelength-stabilized mode-locked thulium-doped fiber laser operating beyond 2 μm ", ELiOs 2021, Lille, France.
- A. Fuerbach, T.T. Fernandez, S. Gross, L. Xu, G. Bharathan: "Femtosecond laser inscribed mid-IR fibre Bragg gratings", invited talk, 9th European Conference on Applications of Femtosecond Lasers in Materials Science (FemtoMat), Mauterndorf Castle, Mauterndorf, Austria (2022).
- A. Fuerbach, T. Fernandez, G. Bharathan, S. Rehman: "Mid-IR photonics", invited talk, Photon-2020, Nottingham, UK (held as a virtual event) (2020).
- A. Fuerbach, G. Bharathan, S. Rehman, L. Xu, T. Fernandez: "Towards all-integrated mid-infrared light sources", invited talk, 22nd International Conference on Transparent Optical Networks (ICTON), Bari, Italy (held as a virtual event) (2020).

- Gayathri Bharathan, Toney Teddy Fernadez, Martin Ams, Jean-Yves Carrée, Samuel Poulain, Marcel Poulain and Alex Fuerbach, "Femtosecond Laser Direct Written Fibre Bragg Gratings in InF3 Fibre", Advanced Photonics Congress 2020, Canada (held as a virtual event).
- Gayathri Bharathan, Xiantao Jiang, Han Zhang, Feng Chen, Ziqi Li, Stuart D. Jackson and Alex Fuerbach, "Mode-locked mid-IR fibre laser based on 2D nanomaterials", Proc. SPIE 11200, AOS Australian Conference on Optical Fibre Technology (ACOFT) and Australian Conference on Optics, Lasers, and Spectroscopy, ANZCOP 2019, Melbourne, Australia.
- Deeksha Beniwal, Huy T. Cao, Sebastian W. S. Ng, Aidan Brooks, Gayathri Bharathan, Alex Fuerbach, Peter J. Veitch and David J. Ottaway "Mid-IR laser for wavefront correction in gravitational wave detectors", Proc. SPIE 11200, AOS Australian Conference on Optical Fibre Technology (ACOFT) and Australian Conference on Optics, Lasers, and Spectroscopy (ACOLS) 2019, ANZCOP 2019, Melbourne, Australia.
- A. Fuerbach, G. Bharathan, T.T. Fernandez, M. Ams: "Fabrication of gratings in mid-infrared compatible fibres via femtosecond laser direct inscription", 21st International Conference on Transparent Optical Networks (ICTON), Angers, France 2019 (Invited Talk).
- Gayathri Bharathan, Toney Teddy Fernandez, Martin Ams, Robert I. Woodward, Darren D. Hudson and Alex Fuerbach, "Direct femtosecond laser inscription of highly reflective and low loss mid-infrared fiber Bragg gratings", CLEO Europe 2019, Munich, Germany.
- R. I. Woodward, M. R. Majewski, G. Bharathan, D. D. Hudson, A. Fuerbach and S. D. Jackson, "Mid-infrared fiber sources for real-time biomedical sensing", SPIE Photonics West 2019, San Francisco, United States (Invited Paper).
- Gayathri Bharathan, "Microstructured in-fibre components for the development of all-fibre mid-infrared laser system", IONS KOALA 2018, Sydney, Australia.
- R. I. Woodward, M. R. Majewski, G. Bharathan, D. D. Hudson, A. Fuerbach and S. D. Jackson "Mid-infrared technologies and opportunities using dysprosium-doped fluoride fiber", IEEE - 2018 the Optoelectronics Global Conference (OGC 2018), Shenzhen, China (Invited Talk).
- R. I. Woodward, M. R. Majewski, G. Bharathan, D. D. Hudson, A. Fuerbach and S. D. Jackson "High-efficiency watt-level mid-infrared fiber laser beyond 3 μm using Dy:ZBLAN", CLEO Pacific Rim 2018, Hong Kong.
- G. Bharathan, R. I. Woodward, D. D. Hudson, S. D. Jackson, and A. Fuerbach "In-Fibre Polarizer for Mid-Infrared Fibre Lasers Based on 45° Tilted Fluoride Fibre Bragg Grating", Advanced Photonics Congress 2018, Zurich, Switzerland.
- A. Fuerbach, G. Bharathan, S. Antipov, M. Ams, R. J. Williams, D. D. Hudson, R. I. Woodward and S. D. Jackson, "Line-by-line Femtosecond FBG Inscription for Innovative Fiber Lasers", Advanced Photonics Congress 2018, Zurich, Switzerland (Invited Talk).
- Thorsten A. Goebel, Gayathri Bharathan, Daniel Richter, Ria Kramer, Martin Ams, Alex Fuerbach and Stefan Nolte, "Aperiodic fiber Bragg gratings written by ultrashort laser pulses using the line-by-line technique", Advanced Photonics Congress 2018, Zurich, Switzerland.
- Thorsten A. Goebel, Gayathri Bharathan, Daniel Richter, Ria Kramer, Martin Ams, Alex Fuerbach and Stefan Nolte, "Ultrashort pulse point-by-point written aperiodic fiber Bragg gratings for suppression of OH-emission lines", SPIE Photonics West 2018.
- G. Bharathan, R. I. Woodward, M. Ams, D. D. Hudson, S. D. Jackson, and A. Fuerbach "Widely Tunable Mid-Infrared All-Fibre Lasers Based on Mechanically Robust Fibre Bragg Gratings", ANZCOP 2017, Queenstown, New Zealand.
- Gayathri B, Dhanesh C V, B. Srinivasan and A. Prabhakar, "Development of 535 nm frequency doubled fiber laser with enhanced spatial coherence for diabetic retinopathy", Workshop on Recent Advances in Photonics, WRAP2015, IISc, Bangalore, India.
- Gayathri B, G Venkat and A. Prabhakar, "The Advanced Photonics Laboratory at IIT Madras", Education and Training in Optics and Photonics, ETOP 2015, Bordeaux, France.

AWARDS, FELLOWSHIPS, AND RECOGNITIONS

- PhD supervisor affiliated with the AUFRANDE (Australia-France Doctoral Network of Excellence) program.
- CLEO US 2023/2024/2025 Technical program committee member and reviewer for the subconference
- Nonlinear Optics.
- Deputy coordinator for the MEFISTA project (research and innovation programme under the Marie Skłodowska-Curie grant agreement), EPFL, Switzerland.
- Global Talent Visa holder in Australia, 2022.
- Nominated for the Bragg Gold Medal (Australian Institute of Physics) for the best Physics PhD in Australia, 2021.
- Best Physics PhD in New South Wales (NSW), Australia, 2021.
- EPFL Postdoctoral Fellowship, EPFL, Switzerland, 2021 (For a period of 4 years).
- Won Australian Optical Society's (AOS's) Postgraduate Student Prize 2019.
- Popular choice award for the poster, Research in Photonics Poster Day 2019, at University of Sydney.
- Won Australian Institute of Physics' (AIP's) Jak Kelly Postgraduate Award 2019.
- Student representative from Macquarie University in Australian Institute of Physics, Physics Postgraduate Awards competition 2019.
- Won first prize, Dept. of Physics and Astronomy Postgraduate Award 2019, at Macquarie University.
- Popular choice award for the poster, Dept. of Physics and Astronomy Postgraduate Research Showcase Day 2018, at Macquarie University.
- Received Postgraduate Research Fund (PGRF 2018) from Macquarie University, Australia.
- BGPP Student Travel Award, Advanced Photonics Congress 2018, Zurich, Switzerland.
- Best paper award, BGPP, Advanced Photonics Congress 2018, Zurich, Switzerland.
- iMQRES Fellowship, Macquarie University, Australia, 2018 (For a period of 3 years).
- Participated in 3MT in Dept. of Physics and Astronomy, Macquarie University, Australia, 2018.
- iMQRTP Fellowship, Macquarie University, Australia, 2017 (For a period of 1 year).
- Official reviewer for the OSA journals - Optics Express, Optics Letters, Optics Continuum and Applied Optics.
- Official reviewer for the IEEE journal - Transactions on Industrial Electronics.
- Official reviewer for the IEEE Photonics journal
- Vice-chair of Optica's (previously OSA) "Fundamental Laser Sciences" Technical Group
- Invited speaker at VIT University, Chennai, 2020.
- Session Chair in VICFCNT-2021, VIT University, Chennai, 2021.
- Guest editor - Special issue on Mid-infrared Laser based Sensors (Sensors, journal by MDPI)

PROFESSIONAL MEMBERSHIP

- Member of Optical Society of America (OSA), now OPTICA
- Member of Australian Optical Society (AOS)
- Member of Society of Photo-Optical Instrumentation Engineers (SPIE)

OUTREACH ACTIVITIES

- Active member of OSA (now OPTICA) student chapter at IIT Madras, and conducted school visits to encourage science activities among students
- Active participant in the organisation of Astronomy Night at Macquarie University in Australia
- Active member of Women in Science at EPFL, Switzerland

FUNDED PROJECTS

NAME OF AGENCY	TITLE OF PROJECT	STATUS
Defence Research and Development Organisation (DRDO), Govt. of India	Development of 2 um and 4 um Laser sources for countermeasure applications	Ongoing
Seed Grant, Indian Institute of Technology (IIT) Delhi.	All-fiber mid-infrared laser source for medical applications	Ongoing
Equipment Matching Grant, Indian Institute of Technology (IIT) Delhi.	Development of 2 um and 4 um Laser sources for countermeasure applications	Ongoing
Anusandhan National Research Foundation (ANRF), Govt. of India.	Development of a true all-fiber mid-infrared supercontinuum source for spectroscopy applications	Proposal Submitted

SUPERVISION DETAILS

NAME OF THE STUDENT	INSTITUTE NAME	DEGREE	THESIS TITLE/ COMMENTS
Khadija Karim	Macquarie University, Sydney, Australia.	Master of Research (MRes)	Development of compact mid-infrared fibre lasers using 2D-nanomaterials and side-polished fibre. (Degree Awarded)
Moritz Bartnick	Ecole Polytechnique Federal de Lausanne (EPFL), Switzerland.	Doctor of Philosophy (PhD)	Thulium-doped fiber cavities for frequency comb generation at 2 micron (Degree Awarded)
Luyi Xu	Macquarie University, Sydney, Australia.	Master of Research (MRes) & Doctor of Philosophy (PhD)	Investigation into the thermal properties of femtosecond laser inscribed flouride fibre Bragg gratings (Thesis Submitted)
Thuy Ha	Macquarie University, Sydney, Australia.	Doctor of Philosophy (PhD)	Chalcogenide-glass waveguide arrays for mid-infrared mode-locked fiber lasers (Thesis Submitted)
Wajahat Hussain	Macquarie University, Sydney, Australia.	Doctor of Philosophy (PhD)	Currently supervising

SUPERVISION DETAILS

NAME OF THE STUDENT	INSTITUTE NAME	DEGREE	THESIS TITLE/ COMMENTS
Sebastian Lukas Lema Alveteg	Macquarie University, Sydney, Australia.	Doctor of Philosophy (PhD)	Currently supervising (under AUFRANDE program)
Sobia Rehman	Macquarie University, Sydney, Australia.	Doctor of Philosophy (PhD)	She is currently a final year PhD student.
Bhanu Yadav	IIT Delhi, India	Doctor of Philosophy (PhD)	Currently supervising
Dr. Sanyogitha Singh	IIT Delhi, India	Post-doctoral fellow (Post-doc)	Currently working as my Post-doc in the DRDO project

PUBLICATIONS & RESEARCH IMPACT

- Over 30 peer-reviewed publications in high-impact journals and conferences, with 450+ citations.
- Google Scholar: <https://scholar.google.com.au/citations?user=C7Oz8iOAAAAJ&hl=en>
- <https://opc.iitd.ac.in/faculty/prof-gayathri-bharathan>

REFERENCES

- References available upon request.